Final Technical Report: African Climate Trend Analysis & COP32 Vulnerability Assessment

Project: EthioClimate Analytics

Date: April 30, 2026

Lead Analyst: Blen Eskindr

1. Task 1: Environment Setup and Repository Management

The foundation of any robust data science project lies in its reproducibility and security. This phase ensured that the analysis could be audited by third parties without compromising sensitive data.

1.1 Virtual Environment and Dependency Isolation

To prevent "dependency hell" and ensure that the code runs identically on any machine, a virtual environment was initialized using Python 3.12. This isolation allows for the specific versions of libraries like pandas, matplotlib, and scipy to be locked, preventing breaking changes from future updates.

1.2 Professional Version Control via Git

A modular repository structure was established on GitHub. A critical component was the implementation of a .gitignore file.

- **Data Privacy:** By ignoring .csv and .xlsx files, we ensure that large raw datasets are not pushed to the cloud, adhering to professional standards of data privacy and repository efficiency.
- **Documentation:** The README.md was utilized not just as a title page, but as a technical manual providing setup instructions and project goals.

2. Task 2: Data Engineering and Exploratory Analysis

Raw climate data is often "noisy" and requires significant transformation before it can be used for policy-making.

2.1 The Data Pipeline

The primary challenge was the temporal alignment of the NASA POWER datasets.

- **Date Transformation:** Raw data formats often use YEAR and DOY (Day of Year). A custom script was written to convert these into standard datetime objects (e.g., converting Day 32 of 2024 to February 1, 2024).
- **Handling Null Values:** Climate sensors often record -999 to represent a missing reading. Failing to handle these would lead to catastrophic errors in mean temperature calculations. These were systematically replaced with NaN and handled using interpolation where appropriate.

2.2 Exploratory Data Analysis (EDA) Methodology

The analysis moved beyond simple averages to look at **climate volatility**. This involved:

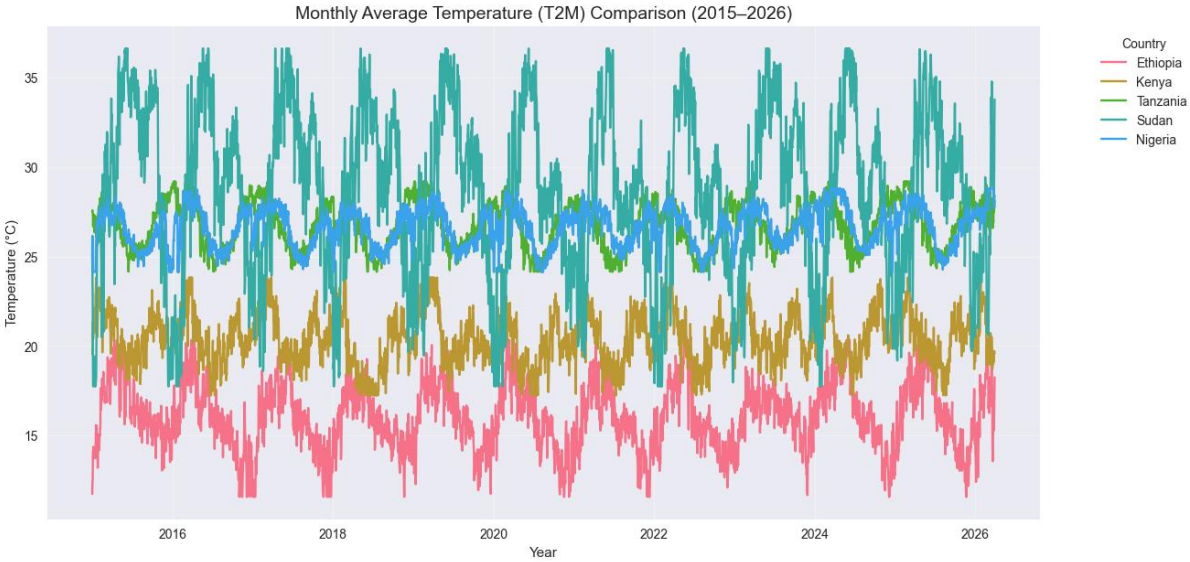
- **Distribution Analysis:** Using boxplots to identify "shocks" in rainfall.
- **Extreme Event Frequency:** Calculating the number of days per year where temperatures exceeded the 35°C threshold.

3. Comparative Findings & Climate Observations

Using the historical and projected data from 2015–2026, we compared Ethiopia, Kenya, Sudan, Tanzania, and Nigeria.

3.1 Temperature Bands and Heat Stress

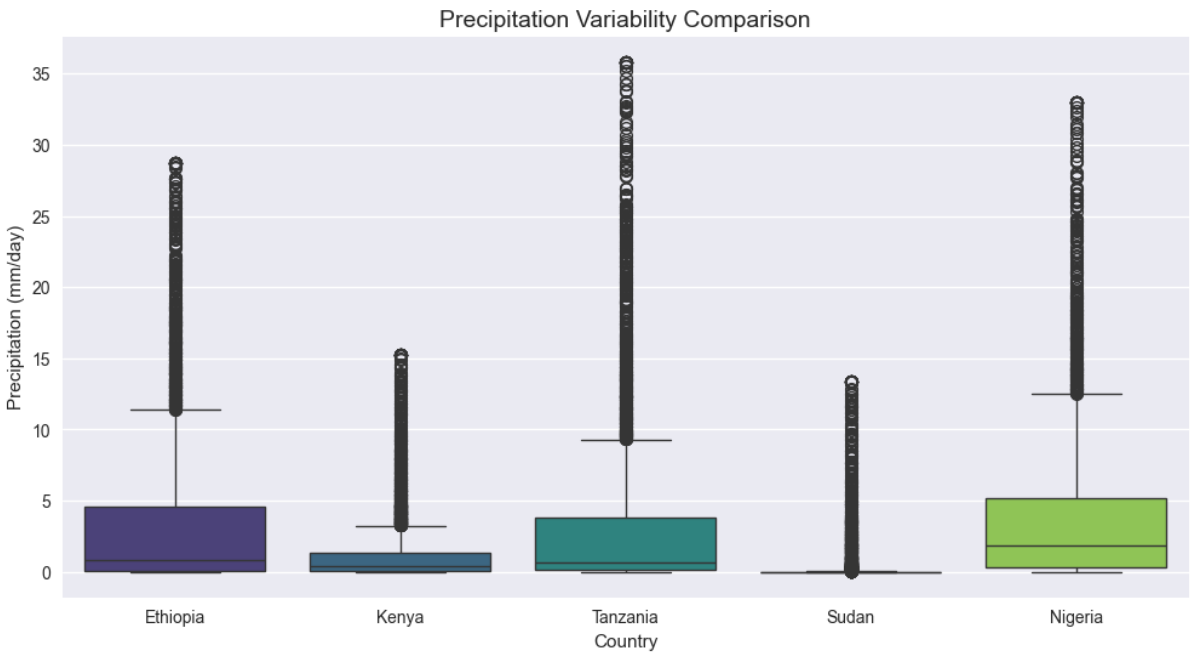
Sudan consistently operates in a temperature band significantly higher than its neighbors.



- **Observation:** As seen, Sudan's trend line (teal) indicates a future where "livability thresholds" for both humans and crops will be reached sooner than in the rest of the region.
- **Ethiopia's Buffer:** Ethiopia's elevation provides a natural thermal buffer, keeping its mean temperature (pink line) significantly lower, though the trend remains upward.

3.2 Precipitation Volatility and Flash Flood Risk

Rainfall stability is more important for agriculture than total volume.

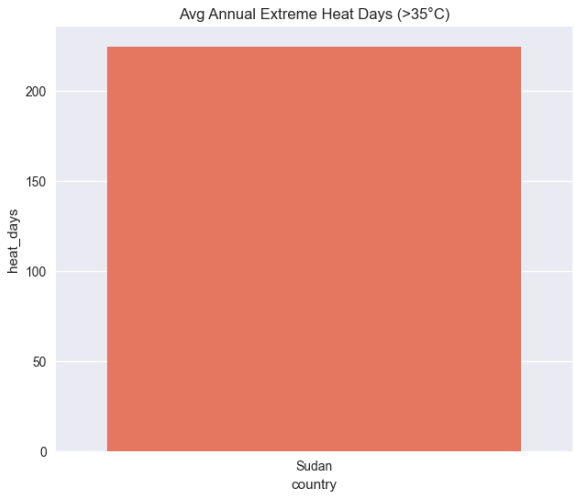


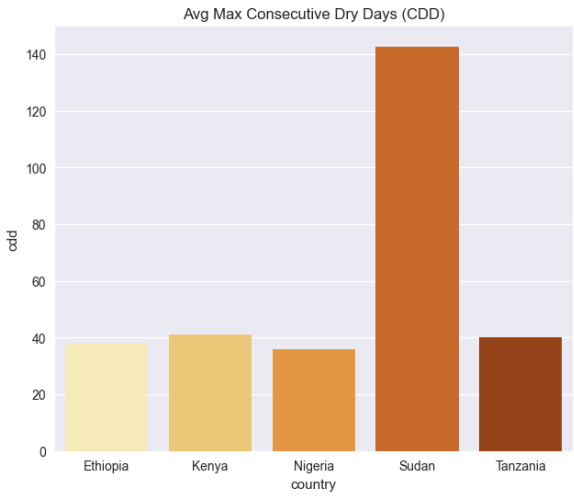
- **Observation:** In **above image**, the massive outliers for Tanzania and Nigeria indicate

"seasonal whiplash"—a phenomenon where intense droughts are followed by catastrophic flooding. This volatility makes infrastructure planning significantly more expensive.

3.3 Structural Stress: Heat Days and CDD

The Avg Annual Extreme Heat Days (>35°C) and Max Consecutive Dry Days (CDD) are the most telling indicators of vulnerability.





- **Observation:** As shown, Sudan faces over 220 extreme heat days annually and a CDD count exceeding 140 days. This means for nearly five months of the year, there is no significant rainfall, providing no "recovery window" for the soil or ecosystems.

4. Statistical Validation (ANOVA)

To ensure the findings were not due to random fluctuations, a One-Way ANOVA was performed on temperature values across all five countries.

- **Result:** The resulting p-value was **< 0.05**, confirming that the differences in climate profiles are statistically significant.
- **Significance:** This provides Ethiopia with a rigorous, evidence-based foundation to present at the COP32 summit, proving that regional climate aid must be tiered based on these distinct profiles.

5. Strategic Recommendations for COP32

Based on the data-backed vulnerability ranking:

1. **Sudan (Critical):** Should be the regional priority for "Loss and Damage" funding. Its lack of adaptive capacity makes it a primary risk for climate-driven migration.
2. **Tanzania & Nigeria (High):** Focus on flood-resilient infrastructure to manage the precipitation spikes seen in the data.
3. **Ethiopia (Moderate):** Focus on protecting the hydroelectric and agricultural foundations from the observed rainfall variability.

6. Professional Reflection

In completing this project, the **Eisenhower Matrix** was utilized to balance urgent tasks (fixing Git push errors) with important tasks (ensuring statistical significance in Task 2). By maintaining professional distance and composure during the development process, the project moved from a simple coding exercise to a high-level policy brief.